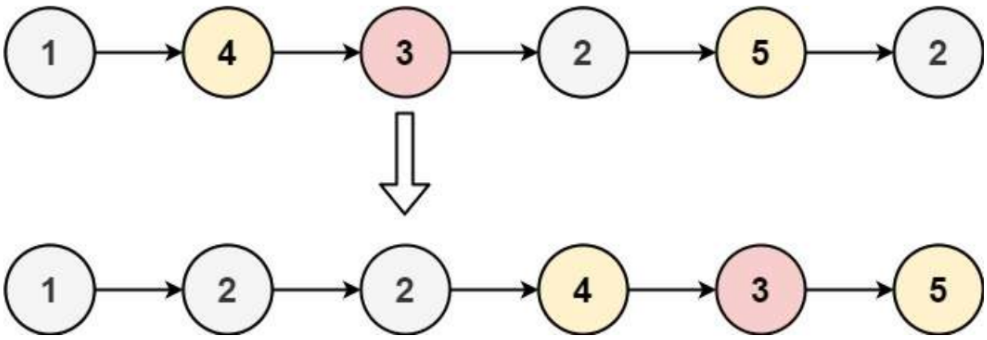
Problem Statement

Given the head of a linked list and a value x, partition it such that all nodes less than x come before nodes greater than or equal x.

You should preserve the original relative order of the nodes in each of the two partitions.

Example 1



Input:
6
1 4 3 2 5 2
3

Output:
1->2->2->4->3->5

Example 2

Input:
2
2 1
2

Output:
1->2

Input Format

The first line of input consists of an integer **N** representing the number of nodes in the linked list.
The second line of input consists of **N** space separated integers representing the elements of the nodes in the linked list.
The third line of input consists of an integer **X** representing the partition value.

Output Format

The output prints the partitioned linked list separated by an arrow **->**.

Refer to the sample output for the formatting specifications.

Constraints

In this scenario, the given test cases will fall under the following constraints:
$2 \leq N \leq 8$
$1 \leq$ Elements of the node ≤ 200
$1 \leq X \leq 200$

Sample Input

```
6
1 4 3 2 5 2
3
```

Sample Output

```
1->2->2->4->3->5
```

Sample Input

```
2
2 1
2
```

Sample Output

```
1->2
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Problem Statement

Given a string **s** consisting of words and spaces, return the length of the last word in the string.

Example 1:
Input: s = "Hello World"
Output:
5
World
Explanation: The last word is "World" with length 5.

Example 2:
Input: s = "luffy is still joyboy"
Output:
6
joyboy
Explanation: The last word is "joyboy" with length 6.

Input Format

The input consists of a string **s.**

Output Format

The first line of output prints the length of the last word in the string.
The second line prints the last word in the string.

Refer to the sample output for the formatting specifications.

Constraints

1 ≤ s.length ≤ 100
s consists of only English letters and spaces ' '.
There will be at least one word in s.

Sample Input **Sample Output**

```
Hello World
```

```
5
World
```

Sample Input **Sample Output**

```
luffy is still joyboy
```

A company is transmitting data to another server. The data is in the form of numbers. To secure the data during transmission, they plan to obtain a security key that will be sent along with the data. The security key is identified as the count of the repeating digits in the data.

Write an algorithm to find the security key for the data.

Input

The input consists of an integer data, representing the data to be transmitted.

Output

Print an integer representing the security key for the given data. If no data is repeated it should display

-1

Constraints

NA

Example

Input:

578378923

Output: 3

Explanation

The repeated digits in the data are 7, 8 and 3. So, the security key is 3.

Public Test Cases

#	Input	Expected Output
1	578378923	3

You are playing an online game. In the game, a list of N numbers is given. The player has to arrange the numbers so that all the odd numbers of the list come after the even numbers. Write an algorithm to arrange the given list such that all the odd numbers of the list come after the even numbers.

Input

The first line of the input consists of an integer num, representing the size of the list (N).

The second line of the input consists of N space-separated integers representing the values of the list.

Output

Print N space-separated integers such that all the odd numbers of the list come after the even numbers.

Constraints

NA

Example

Input :

8

10 98 3 33 12 22 21 11

Output:

10 98 12 22 3 33 21 11

Explanation

All the even numbers are placed before all the odd numbers.

Public Test Cases

#	Input	Expected Output
1	8 10 98 3 33 12 22 21 11	10 98 12 22 3 33 21 11

An e-commerce site wishes to enhance its ordering process. They plan to implement a new scheme of OTP (One Time Password) generation for order confirmations. The OTP can be any number of digits. For OTP generation, the user will be asked to enter two random numbers. The first number entered should always be smaller than the second number. The OTP is calculated as the sum of the maximum and the minimum prime values in the range of the user-entered numbers.

Write an algorithm to find the OTP

Input

The input consists of two space-separated integers – firstNumber and secondNumber, representing the two numbers entered by the user. Both numbers are considered in the range.

Output



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Print an integer representing the sum of the largest and smallest prime numbers in the range of given numbers.

Constraints

-109< firstNumber <secondNumbers<109

Example

Input:

-97 50

Output:

50

Explanation

The smallest and largest prime numbers within the given numbers are -97 and 47, respectively. The sum of -97 and 47 is 50. So, the output is 50.

Public Test Cases

#	Input	Expected Output
1	-97 50	50

1). How to find first repeated character of a given String?

Examples:

Input : "I gave an apple"
Output : 'a'
Input : "Hold me high"
Output : 'h'

3). Find the nearest square number of the array element for the given position and store it in the start of the array & sort the remaining elements of the array. **Marks: 10**

Example:

Input: {45, 23, 11, 57, 68, 32} & Position: 2 (Here position 2 refers to number 23)
Output: {25, 11, 32, 45, 57, 68}

Input: {45, 23, 11, 57, 68, 32} & Position: 4
Output: {64, 11, 23, 32, 45, 68}

4). Write a function that given a list of non negative integers, arranges them such that they form the largest possible number.

Marks:15

Example:
Input : [50, 2, 1, 9]
Output: 95021

5). Given multiple ranges [L, R] and a prime number p, we need to find all P-smooth numbers in given individual ranges.
(An integer is P – smooth number if the largest Prime factor of that number <= p.
Marks:

Examples:

Input :
p = 7
ranges[] = {[1, 17], [10, 25]}

Output :
For first range : 1 2 3 4 5 6 7 8 9 10 12 14 15 16
For second range : 10 15 16 18 20 21 24 25

6). Given an array of N numbers, we need to maximize the sum of selected numbers. At eac step you need to select a number Ai, delete one occurrence of Ai-1 (if exists), Ai+1 (if exist and Ai each from the array. Repeat these steps until the array gets empty. The problem is to maximize the sum of selected numbers.
Note: We have to delete Ai+1 and Ai-1 elements if they are present in the array and not Ai- Ai-1.
Marks:

Examples:

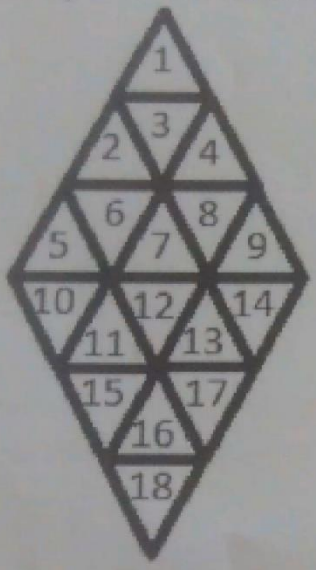
Input : a[] = {1, 2, 3}
Output : 4

Explanation: At first step we select 1, so 1 and 2 are deleted from the sequence leaving us Then we select 3 from the sequence and delete it.So the sum of selected numbers is 1+3 =

Input : a[] = {1, 2, 2, 2, 3, 4}
Output : 10

Explanation : Select one of the 2's from the array, so 2, 2-1, 2+1 will be deleted and we with {2, 2, 4}, since 1 and 3 are deleted. Select 2 in next two steps, and then select 4 in step. We get a sum of 2+2+2+4=10 which is the maximum possible.

8). Given a triangular assortment of size N where the total number of rows is N, which is always even. The triangles are numbered starting with the triangle in the first row and increasing left to right and top to bottom. An example of such an arrangement with N=6 is given below.

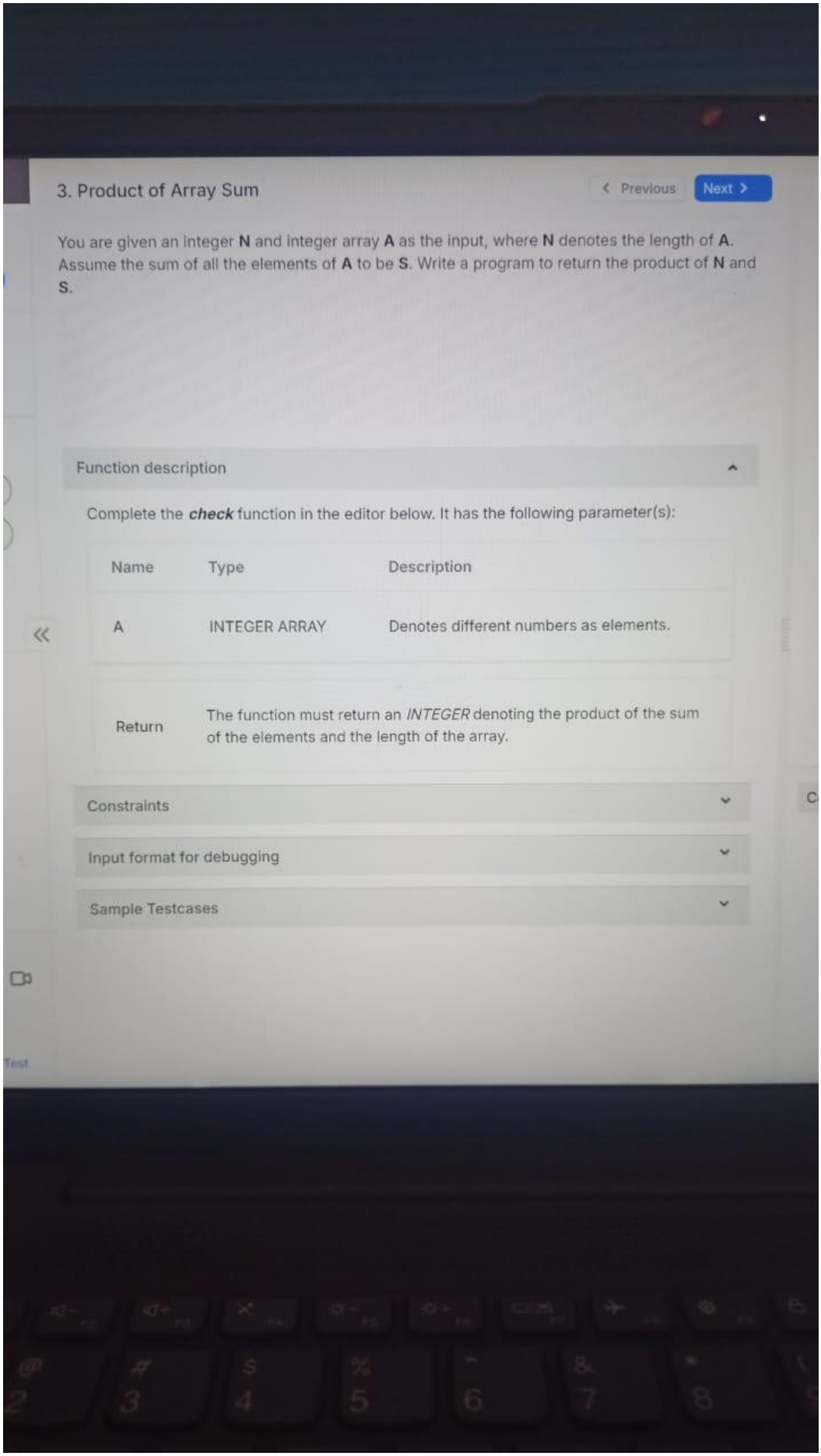


The task is to find minimum distance D from a starting triangle to all the other triangles. The distance between two triangles is the minimum number of edges to be crossed to reach the destination triangle from the starting triangle. The distance between any triangle to its adjacent triangle is 1 unit.

Marks:25

Input : 6 8

Output : 3 3 2 1 3 2 1 0 1 4 3 2 3 2 4 5 4 6



3. Product of Array Sum

Previous Next

You are given an integer **N** and integer array **A** as the input, where **N** denotes the length of **A**. Assume the sum of all the elements of **A** to be **S**. Write a program to return the product of **N** and **S**.

Function description

Complete the ***check*** function in the editor below. It has the following parameter(s):

Name	Type	Description
A	INTEGER ARRAY	Denotes different numbers as elements.

Return	The function must return an *INTEGER* denoting the product of the sum of the elements and the length of the array.

Constraints

Input format for debugging

Sample Testcases

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Coding

Revisit Later

Question 2

How to Attempt?

Count the Visible Clothes

Aunt Jenny recently hired a lady to look after her household work so that she could concentrate on setting up her new business. As part of the work, the lady is supposed to clean the house, wash the clothes, and hang them up on a clothesline every day. However, there is a problem with the drying of the clothes as the clothesline is too short to accommodate all the clothes. So, the lady places one cloth on top of the other. Due to this, some of the clothes are covered partially and some are covered completely.

Consider a clothesline of length N meters divided into N equal sections extending from 0 to N. The order in which the input is received is the order in which the clothes are hung on the clothesline.
Given the order and the position in which the clothes are hung, your task is to return the number of clothes that are visible (partially or completely) when seen from the front.

Note: *Ignore the other dimensions of the clothes for the purpose of this problem.*

Input Specification:

input1: An integer value N denoting the length of the clothesline (1 <= N <= 10000)

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Coding

Revisit Later

Question 2

How to Attempt?

Count the Visible Clothes

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Note: *Ignore the other dimensions of the clothes for the purpose of this problem.*

Input Specification:

input1: An integer value N denoting the length of the clothesline (1 <= N <= 10000)

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1. Coding

input1: An integer value N denoting the length of the clothesline (1 <= N <= 10000)
input2: An integer value representing the number of clothes placed on the clothesline (1 <= input2 <= 10000)
input3: A 2D integer array of size input2*2 giving the start and ending positions of clothes that were hung

Output Specification:

Return the count of clothes visible completely or partially from the front.

Example 1:

input1: 10
input2: 5
input3: { {0,4}, {6,3}, {1,5}, {6,4}, {7,2} }

Output: 4

Explanation:
Here N=10 which is the length of the rope in meters and 5 is the number of clothes hung on a rope.
The first cloth starts from 0 and covers 4 sections. Similarly, the second cloth starts at 6 and covers 3 sections from 6 to 9, and so on for the rest of the clothing.
Hence from the below diagram we know that the total number of clothes visible when seen from the front is 4. Therefore, 4 is returned as the final output.

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